

Irrationality of $\sqrt{2}$

Theorem 1. $\sqrt{2}$ is irrational; that is, there is no rational number r such that $r^2 = 2$.

Proof. We first record two elementary facts that will be used in the proof.

First, every rational number has a representation with no common divisor greater than 1. Indeed, let $r \in \mathbb{Q}$. By definition, there exist integers a, b with $b \neq 0$ such that

$$r = \frac{a}{b}.$$

Replacing both a and b by their negatives if necessary, we may assume $b > 0$. Consider the set

$$D = \{s \in \mathbb{Z}_{>0} : \text{there exists } t \in \mathbb{Z} \text{ such that } r = \frac{t}{s}\}.$$

This set is nonempty, since $b \in D$. By the well-ordering principle, D has a least element; call it q . Choose $p \in \mathbb{Z}$ such that

$$r = \frac{p}{q}.$$

We claim that no integer $d > 1$ divides both p and q . If such a d existed, then $p = dp'$ and $q = dq'$ for some integers p', q' , with $q' > 0$. Then

$$r = \frac{p}{q} = \frac{dp'}{dq'} = \frac{p'}{q'},$$

and since $d > 1$, we have $0 < q' < q$. This contradicts the minimality of q . Hence this representation satisfies $\gcd(p, q) = 1$.

Second, we establish the parity lemma: if n^2 is even, then n is even. By the division algorithm with divisor 2, every integer n can be written in exactly one of the two forms

$$n = 2m \quad \text{or} \quad n = 2m + 1$$

for some integer m . If $n = 2m$, then n is even. If instead $n = 2m + 1$, then

$$n^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1,$$

which is odd. Therefore, since every integer is either even or odd (exclusively), an integer whose square is even must itself be even.

Now suppose, for contradiction, that there exists a rational number r such that $r^2 = 2$. By the reduced-representation fact proved above, choose integers

p, q with $q > 0$ such that $r = p/q$ and $\gcd(p, q) = 1$. Substituting into $r^2 = 2$ gives

$$\left(\frac{p}{q}\right)^2 = 2.$$

Multiplying by q^2 yields

$$p^2 = 2q^2.$$

Thus p^2 is even. By the parity lemma, p is even, so there exists $k \in \mathbb{Z}$ such that $p = 2k$. Substituting,

$$4k^2 = 2q^2,$$

so $2k^2 = q^2$. Hence q^2 is even. By the same parity lemma, q is even.

Thus 2 divides both p and q , contradicting $\gcd(p, q) = 1$. Therefore no rational number r satisfies $r^2 = 2$, and hence $\sqrt{2}$ is irrational. $\square$